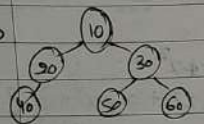
UNIT-4

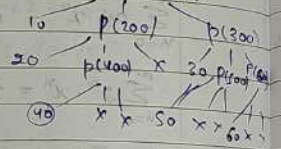
Graphs

⇒ 

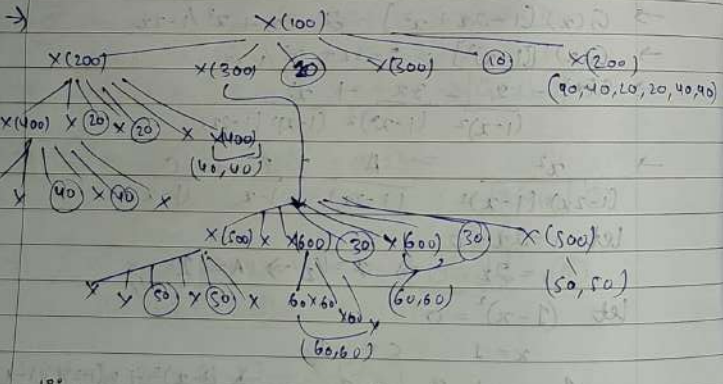
pre - Root left right (10 20 40 30 50 60 — Pre
post - left right root (40 20 10 50 30 60 — Post
In - left root right (40 20 50 60 30 10 — In

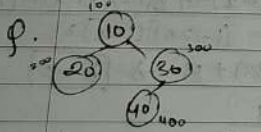
⇒ left → Right → 40, 40, 20, 20, 40, 40, 50, 50, 60, 60, 30
root → right → 60, 60, 30, 50, 50, 10, 50, 50, 60, 60
root → left 30, 60, 60, 30, 50, 50, 10, 40, 40, 20, 20
40 40.

```
Void preorder(Struct node * root)
& if (root != NULL)
  & p.f ("%d", root->info);
    pre(root -> left); }
    pre(root - right); }
```



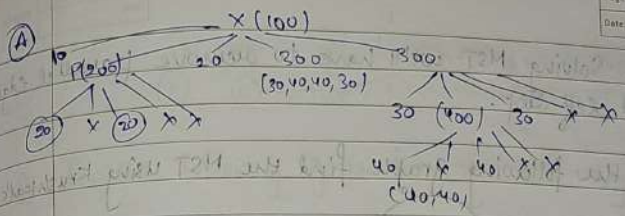
→ 10 20 40 30 50 60

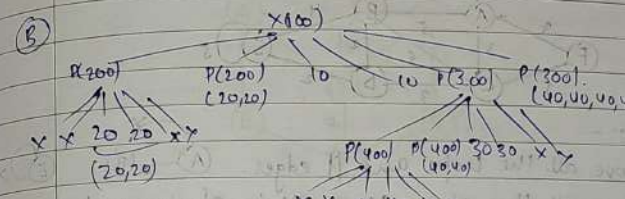
⇒ 

Q. 

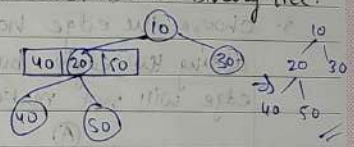
A → 10, 20, 20, 20, 30, 40, 40, 30, 30, 40, 40, 30
B → 20, 20, 20, 20, 10, 10, 40, 40, 40, 40, 30, 30
40, 40, 40, 40, 30, 30.

A: Root left Root Right Right.
B: left left Root Root Right Right.

(A) 

(B) 

Q: Given a Preorder & Inorder construct a binary tree.
In: 40 20 50 10 30
pre: 10 20 40 50 30.



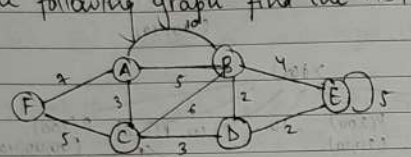
Q. Given only Pre & Post order construct a binary tree.
Inorder of BST is always in increasing order.
pre :-

* Spanning Tree :-

1. Removing 1 edge from spanning tree will make it dis-connected.
2. Adding 1 edge to spanning tree will create a loop.
3. If each edge has a distinct weight then there will be only one unique MST (min spanning tree)
4. Complete ~~undirecting~~ undirecting graph can have n^{n-2} spanning tree where, n is no. of vertices.
5. Disconnect graph doesn't have spanning tree.

(6) for Solving MST we'll have to remove parallel edges and self loop.

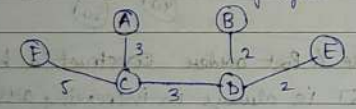
* Q. for the following graph find the MST using Krushkal's Algo.



Steps:—

1. Remove all the loops and || edges.

2. Arrange all the edges acc. to their edge weight in inc order.
   B-D : 2, D-E : 2, A-C : 3, C-D 3, B-E : 4, (A-B : 5), F-C : 5, C-B : 6, F-A : 7, not incl.
3. Choose the edge having min edge weight and keep adding the edges but make sure that adding the edge will not make any cycle.



→ MST = 15.

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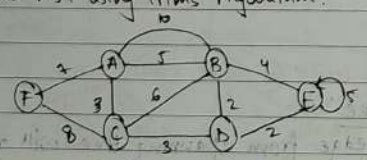
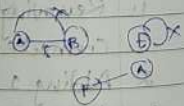
Q. find the MST using Prim's Algorithm.

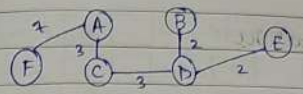


Steps:- (1) Remove all loops and parallel edges.



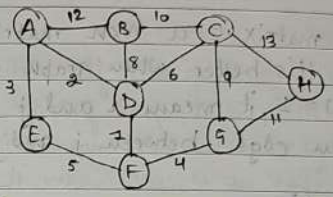
(2) Starting with F {FA : (7), FC (8)}
Starting with A {AC(3), AB(5), FC(8)}
Starting with C {CB(6), CD(3), AB(5), FC(8)}
Starting with D {DB(2), DE(2), CB(6), AB(5), FC(8)}
Starting with B {BE(4), DE(2), CB(6), AB(5), FC(8)}

MST = 17

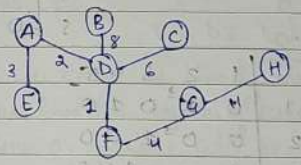


7+3+3+2+2 = 17

Q. for the following graph find the MST using both Algo.

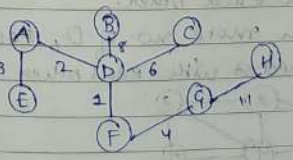


Kruskal's Algo:- D-F : 1 ; A-D : 2, A-E : 3, F-G : 4, E-F : 5, D-C : 6, B-D : 8, C-G : 9, B-C : 10, G-H : 11, A-B : 12, C-H : 13.



MST = 35

Prim's Algo:- (1) Starting with A {AE(3), AD(2), AB (12)}
Starting with D {DF (1), DC (6), DB (8), AE(3), AB(12)}
Starting with F {FE(5), FG(4), DC (6), DB (8), AE(3), AB(12)}
Starting with E {AB(12), FE(5), FG(4), DC(6), DB(8)}
Starting with G {AB(12), FE(5), DC(6), DB(8), GC(9), GH(11)}
Starting with C {AB(12), DB(8), FE(5), GC(9), GH(11), CH(13), CB(10)}
Starting with B {AB(12), GH(11), CH(13), CB(10), BD(8)}
Starting with D {AB(12), GH(11), CH(13), CB(10), DF(1) (done), DC(6), DA(2)}



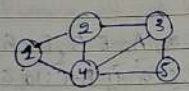
MST = 35

* Graph Representation

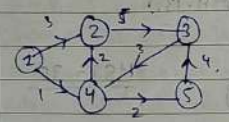
1. Adjacency Matrix
2. Adjacency list
3. Incidence matrix

→ Adjacency matrix is a $n \times n$ matrix where n is no. of vertex. It's better when graph is dense.

→ If $a[i][j] = 1$, it means i and j are adjacent or there is an edge between i and j.



	1	2	3	4	5
1	0	1	0	1	0
2	1	0	1	1	0
3	0	1	0	1	1
4	1	1	1	0	1
5	0	0	1	1	0



	1	2	3	4	5
1	0	1	0	1	0
2	0	0	1	0	0
3	0	0	0	1	0
4	0	1	0	0	1
5	0	0	1	0	0

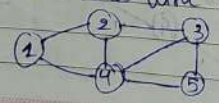
→ Adjacency list, for each vertex a separate linked list is maintained.

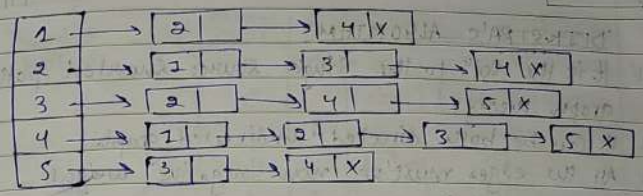
→ It's better when graph is sparse.

→ Dense graph: almost each node is connected with each other.

→ Sparse graph: when more no of 0, each node is not connected with each other.

for eg →

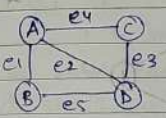




→ Incidence Matrix, If GBA graph of n vertices and e no. of edges and no self loop. It is a $n \times e$ matrix.

→ If $a_{ij} = 1$ it means there is a edge b/w i & j.

If $a_{ij} = 0$ it means there's no edge b/w i & j.



	e1	e2	e3	e4	e5	degree
A	1	1	0	1	0	3
B	1	0	0	0	1	2
C	0	0	1	1	0	2
D	0	1	1	0	1	3
	2	2	2	2	2	10

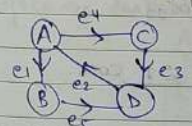
→ It is also called as binary matrix or 0,1 matrix.

→ we can find the degree of vertex using incidence matrix.

→ The sum of columns will be exactly equal to 2.

→ Reduced incidence matrix, to manage space complexity we can remove any node.

→ Outgoing edge means +1, incoming edge means −1 and sum of column 0.



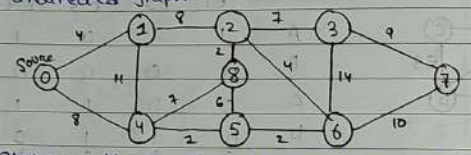
	e1	e2	e3	e4	e5
A	+1	−1	0	+1	0
B	−1	0	0	0	+1
C	0	0	+1	−1	0
D	0	+1	−1	0	−1
	0	0	0	0	0

***

DIJKSTRA'S ALGORITHM

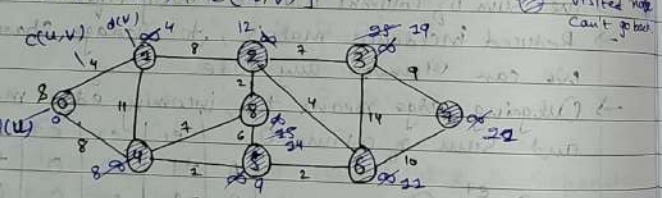
1. It is the sol^n to the single source shortest path in graph theory.
2. It can be both directed & undirected graph.
3. All the edges must've non-negative weight.
4. Graph must be connected
5. Remove all the self loop & parallel edges
6. It can be applied on the weighted graph.

Q. find the shortest path from the source vertex 0 to 7.
undirected graph & cost.



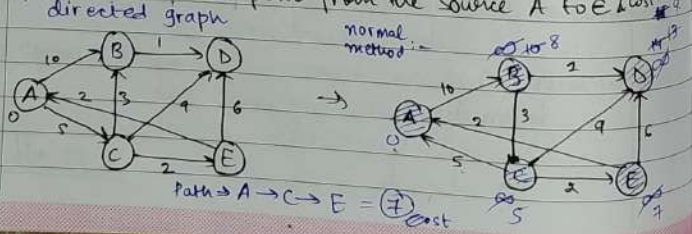
→ Step 1 :- Mark all nodes ∞ except the source node.

★ If $(d(u) + c(u,v) < d(v))$ {
$d(v) = d(u) + c(u,v)$ }



Path → 0→4→5→6→7 = 21 cost

Q. find the shortest path from the source A to E & cost?
directed graph



Path → A→C→E = (7) cost

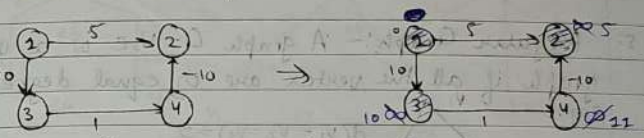
Tabular Method :

Selected vertex	A	B	C	D	E	for Path
A	0	∞	∞	∞	∞	(∞ > 5) ✓
C		10	5	∞	∞	(18 > 8) ✓
E		8		14	7	(8 ≯ 8) ×
B		8		13		(13 > 9) ✓
D				9		

DBCA (reverse)

ACBD = 9 (Path of A–D).

Q.



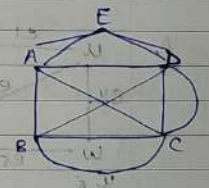
- It may or maynot work for the -ve weight.
- If the total edge weight of the cycle is -ve the definetely dijkstra's algo won't work.

* TYPES OF GRAPH :-

1. Simple graph : It's a graph that can neither the self loop nor the parallel edges.
2. Multi Graph : A graph G = V, E is called multi graph, if it has ||^l edges but no self loop.

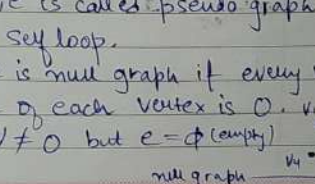
↳ Representation of multi-graph :-
It can only be represented using adjecency matrix.

	A	B	C	D	E
A	0	1	1	1	1
B	1	0	2	1	0
C	1	2	0	2	0
D	1	1	2	0	1
E	1	0	0	1	0

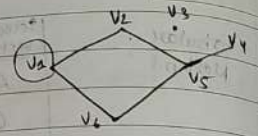


3. Pseudo Graph : A graph G = V, E is called pseudo graph if it has ||^l edges as well as self loop.
4. NULL Graph : A graph G = V, E is null graph if every vertex of G is isolated i.e. degree of each vertex is 0.
$d(v) = 0 \ \forall \ v \in V$, i.e. $V \neq 0$ but $E = \phi$ (empty)

null graph



Degree of Vertex (d(V))



$d(V_1) = 4$ (self loop counted twice)

$d(V_2) = 2$ [$d(V_3) = 0$

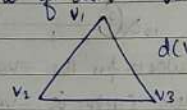
$d(V_4) = 1$ $d(V_5) = 3$. $d(V_6) = 2$

Isolated Vertex : if degree is 0.

Pendent Vertex : the vertex with degree 1. (e.g. V_4)

5. **Regular Graph** :- A graph $G = V, e$ is a regular graph if all the vertex are of equal degree.

e.g.

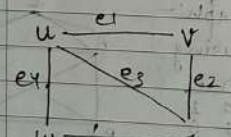


$d(V_1 = V_2 = V_3) = 2$

A Graph h is said to be a sub graph of graph G if all the vertices and edges of h are present in G and each edge of h has same end vertices in h (sub graph) as in G.

$h = (V', e')$ is a subgraph of $G = (V, e)$ if $V' \subseteq V$ and $e' \subseteq e$.

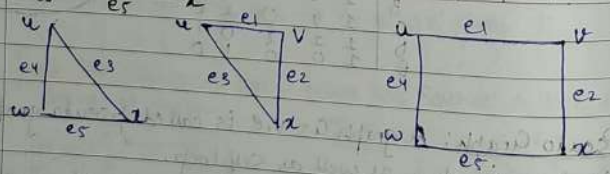
Ex



$G = (V, e)$

$V = \{u, v, w, x\}$

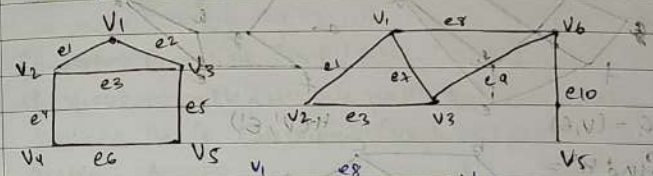
$e = \{e_1, e_2, e_3, e_4, e_5\}$



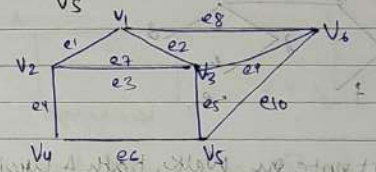
Operations of Graph

1) **Union**

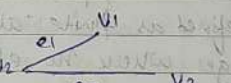
If $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are two graphs then their union $G_1 \cup G_2$ is another graph where $V = V_1 \cup V_2$ and $E = E_1 \cup E_2$.



$G_1 \cup G_2$



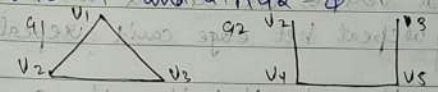
$G_1 \cap G_2$



2) **Intersection**

If $G_1 = (V_1, E_1)$ & $G_2 = (V_2, E_2)$ their intersection $G_1 \cap G_2$ is another graph where $V = V_1 \cap V_2$ & $E = E_1 \cap E_2$

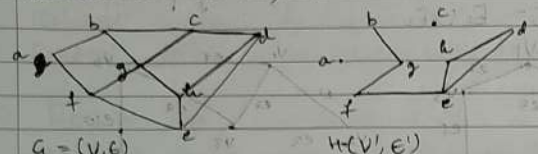
→ A Graph G is said to be decomposed in two sub-graphs if $G_1 \cup G_2 = G$ and $G_1 \cap G_2 = \phi$



$G_1 \cup G_2 =$

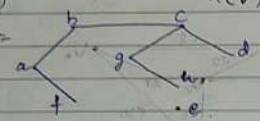
$G_1 \cap G_2 = \phi$

→ **Complement of Sub-Graph:** Let $G=(V,E)$ be any graph and $H=(V',E')$ be it's sub graph then the complement of sub graph w.r.t G is represented by $H^c = G-H = (V'', E'')$ where $E'' = E-E'$, $V'' = V-V'$.



$G=(V,E)$ $H=(V',E')$

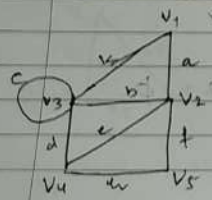
find H^c =



★★Q. Write a short note on Walk, path & Circuit.

- **Walk** A walk is defined as finite alternating sequence of vertices and edges where no edge appear more than once in a walk but vertex can appear more than once. A walk is also called as a chain or edge train. A walk which begin and end with same edge is called as closed walked i.e. they've same terminal vertex and a walk which is not closed that is there terminal vertices are distinct are called open walk.

* Vertex may repeat but edge can't repeat.



$V_1 \, k \, V_3 \, c \, V_3 \, b \, V_2 \, a \, V_1$ (closed)

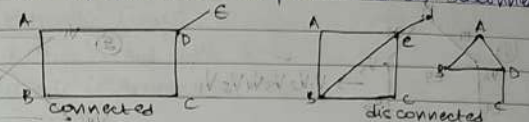
$V_1 \, k \, V_3 \, b \, V_2 \, f \, V_5$ (open)

$V_1 \, k \, V_3 \, b \, V_2 \, f \, V_5 \, h \, V_4$ (path)

or $V_3 \, c \, V_3$ (Circuit).

- **Path:** A path is a open walk where no vertex appear more that once and a path doesn't intersect themselves.
- **Circuit:** It is a closed walk in which no vertex except the initial and final vertex appear more than once. A circuit is a closed non intersecting walk. It is also called as cycle or a circular path.

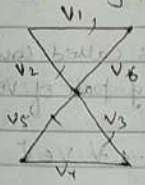
⇒ **Connected Graph:** $G=(V,E)$ is said to be connected if there is atleast one path between every pair of vertices in G, Otherwise Graph is called as disconnected graph.



★★ **Euler Path and Hamiltonian Path**

① A Path in a graph is said to be a euler path if it traverse each edge in the graph once and only once. and a closed walk in a graph which traverse each edge of the graph exactly one is called as euler circuit or euler line and a graph which contains euler line is a euler graph.

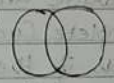
(1)



$V_1 V_2 V_5 V_4 V_3 V_6$

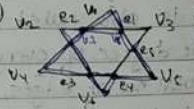
It's a euler graph.

(2)



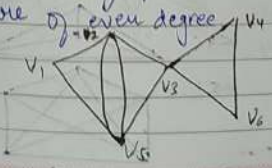
It's a euler graph

(3)



It's a euler graph.

→ A connected graph is a euler graph if all the vertex of graph G are of even degree.

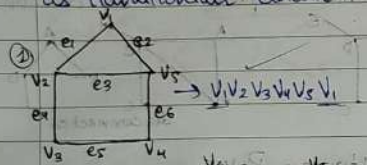


$V_2 \, V_1 \, V_5 \, V_2 \, V_5 \, V_3 \, V_4 \, V_6 \, V_3 \, V_2$

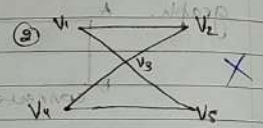
degree $V_1 = 2$, $V_2 = 4$, $V_3 = 4$, $V_4 = 2$, $V_5 = 2$, $V_6 = 2$

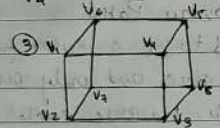
→ Euler Graph is always connected.
→ Euler graph does not contain any isolated or pendant vertex.
→ Euler graph can be decomposed into circuit.

② A path which contain every vertex of graph G exactly once, except the starting and ending vertex is called as hamiltonian path. And a circuit that passes through each vertex of the graph exactly once is called as Hamiltonian Circuit.



→ $v_1 v_2 v_3 v_4 v_5 v_1$





→ $v_1 v_2 v_3 v_8 v_7 v_6 v_5 v_4 v_1$

30/9/25

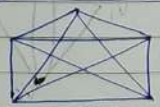
SNP Q. Explain Complete Graph, Bipartitic Graph, Complete Bipartitic Graph.

① Complete Graph: A graph $G=(V,E)$ is called complete graph if there is an edge b/w every pair of vertices. It is also called universal graph.
The degree of every vertices is $d(v) = n-1 \ \forall \ v \in V$
Total no of edges, $E = \frac{n(n-1)}{2}$
It's denoted by K_n, where n stands for no of vertices.
Complete Graph with 5 vertices is called Kurtowski first Graph.

K_2 :

K_3 :

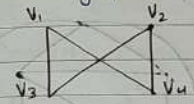
K_5 :



A regular graph of degree k is k-regular where every vertex have 0

② Bipartitic Graph: A graph $G=(V,E)$ is said to be bipartitic if there is a partition (V_1, V_2) of V i.e., $V = V_1 \cup V_2$ and $V_1 \cap V_2 = \phi$ where $V_1 \neq \phi$, $V_2 \neq \phi$
Such that no 2 vertices of V_1 are adjacent and no 2 vertices of V_2 are adjacent to eachother also each edge of graph G connects a vertex of V to a vertex of V_2

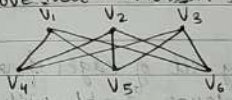
③ Complete Bipartitic Graph: A Graph $G=(V,E)$ is said to be complete if $v_1 \in V_1$ and $v_2 \in V_2$ if $|V_1| = m$ & $|V_2| = n$ then complete bipartitic graph is denoted by $K_{m,n}$



$K(2,2)$

$V_1 = \{v_1, v_2\}$ $V_2 = \{v_3, v_4\}$
$V_1 \cup V_2 = V = \{v_1, v_2, v_3, v_4\}$
$V_1 \cap V_2 = \phi$

Q. Prove that Kurtowski second graph is C.B.G.

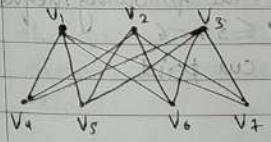


$K_{3,3}$

$V_1 = \{v_1, v_2, v_3\}$ $V_2 = \{v_4, v_5, v_6\}$
$V_1 \cup V_2 = V = \{v_1, v_2, v_3, v_4, v_5, v_6\}$
$V_1 \cap V_2 = \phi$

$K_{3,3}$ graph is also called as Kurtowski Second graph.

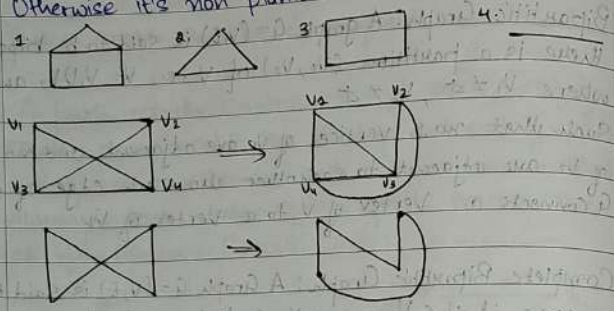
- A Complete Bipartitic Graph $K_{m,n}$ has $m \times n$ edges.
- max no of edges in a C.B.G of n vertices is $n^2/4$.



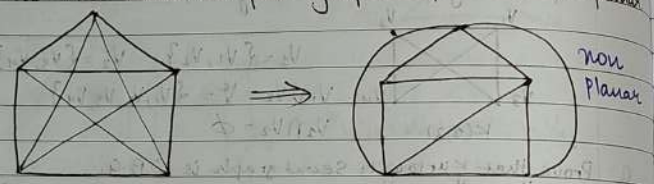
$K_{3,4}$

$V_1 = \{v_1, v_2, v_3\}$
$V_2 = \{v_4, v_5, v_6, v_7\}$
$V_1 \cup V_2 = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$
$V_1 \cap V_2 = \phi$

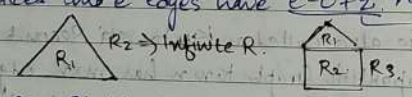
· **Planar Graph:** A graph G is planar if it can be drawn on a plane without cross-over edge. Otherwise it's non planar.



Q. Prove that Kurtowski first graph is planar / non-planar



⇒ **Regions:** A region is defined by set of edges forming it's boundary. A connected planar graph with n vertices and e edges have $e-0+2$ region. euler formula



In any simple connected planar graph with f regions, n vertices and e edges ($e>2$) the following inequalities must hold (1) $e \geq 3/2\, f$ (2) $e \leq 3n-6$.

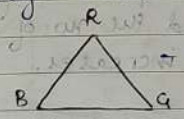
→ Bridge, cutset and curdles cut point.

★ · **Graph Coloring:** —

∴ A coloring of a graph G is an assignment of colours to its vertices so that no two adjacent vertices have the same color.

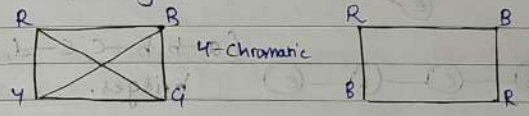
· If they require min no. of colours for coloring of a graph then it's called as proper coloring.

· A graph G that requires K different color for it's proper coloring then its called as K-chromatic graph and the no. K is called as chromatic no. and is denoted by $\chi(G)$.

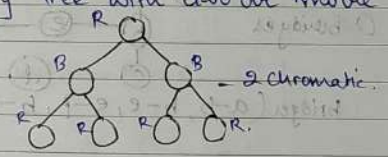


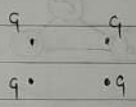
→ 3-chromatic graph

· A complete graph with 'n' vertices is n-chromatic.



· Every Tree with two or more vertices is 2-chromatic



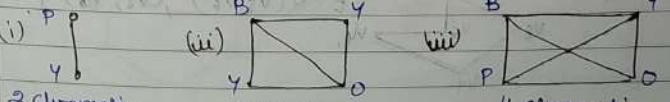


1-chromatic

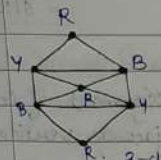
· A graph consisting of only isolated vertex is 1-chromatic.

· If graph is a circuit with n-vertices then it's 2-chromatic if n is even and is 3 chromatic if n is odd.

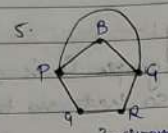
Q. (i)



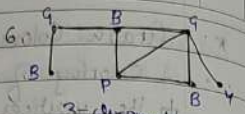
2 chromatic. 3 chromatic 4 chromatic

4. 

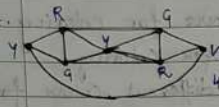
3-chromatic

5. 

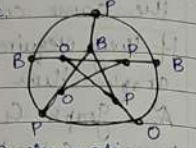
3-chromatic



3-chromatic

7. 

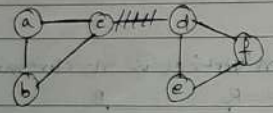
4-chromatic

8. 

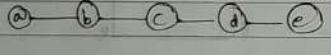
3-chromatic

- **BRIDGES:** —

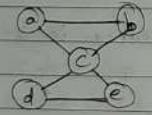
If an edge (u,v) is removed from the graph then graph gets disconnected and the no. of connected component in a graph increases.

(1) 

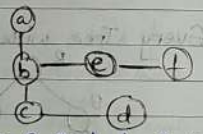
C — D is a bridge.

(2) 

(a-b, b-c, c-d, d-e) bridges.

(3) 

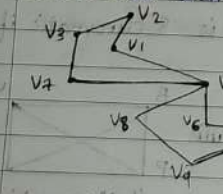
0 bridges

(4) 

bridges (a-b, b-e, e-f, b-c, c-d)

- **CUTSET:** —

A Cutset for a graph G is a smallest set of edges such that removal of the set disconnects the graph.

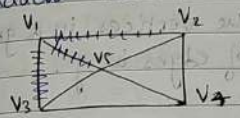


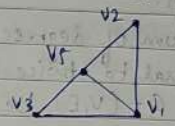
$E = \{(V_1, V_5)\ (V_7, V_5)\}$

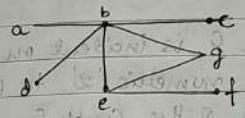
$E = \{(V_5, V_6)\ , (V_5, V_8)\}$

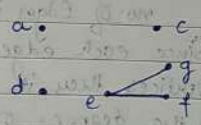
- **CUT POINT / CUT VERTICES:** —

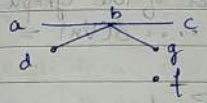
A Cut point for a graph G is a vertex V such that G-V has more connected components than G or disconnected the sub graph G-V is obtained by deleting the vertex V and also deleting all the edges incident on V.

(1) 

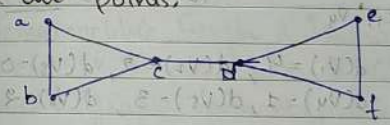
G-V = 

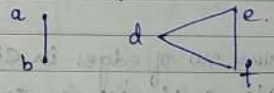
(2) 

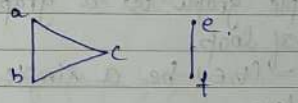
G-V = 



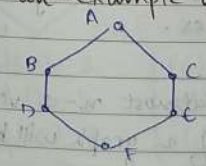
Q. Give an example of a graph with 6 vertices and exactly 2 cut points.



If we remove c, 

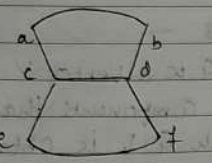
If we remove d, 

Q. Give an example of a graph with 6 vertices and 0 cutpoint



0 cut points

Q.



() cut points.

Proof: —

1. The Sum of degree of all the vertices in a graph g is equal to twice the no. of edges in g.

⇒ let $G = (V, E)$ be a graph

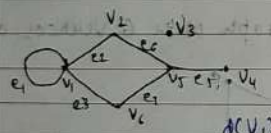
no. of Edges in G is $|E|$.

Since each edge in a graph G is incident on two vertices then it contributes numeric '2' to the sum of the degrees of the vertices of the Graph G.

i.e. $d(V_1) + d(V_2) + d(V_3) + \dots d(V_n) = 2|E|$.

$\Rightarrow \sum_{V \in V} d(V) = 2|E|$

Hence, Proved //



$= 2|E| = 2 \times 6 = \underline{12}$

$d(V_1) = 4$, $d(V_2) = 2$, $d(V_3) = 0$

$d(V_4) = 1$, $d(V_5) = 3$, $d(V_6) = 2$ $= \underline{12}$

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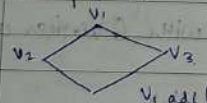
2. Maximum no. of edges in simple graph is $E = n(n-1)/2$.

⇒ • A simple graph is a graph with no parallel edges and self loop.

• Let $G = (V, E)$ be a simple graph with n vertices i.e. $|V| = n$



V_1 adj (V_2, V_4)

• Each vertex is adjacent to n-1 vertices.

• The degree of each vertex is atmost n-1 vertices.

• Sum of degree of vertex of a graph will be $n*(n-1)$

$\sum_{V \in V} d(V) = 2|E| \Rightarrow n*(n-1) = 2|E|$

$\therefore |E| = \frac{n(n-1)}{2}$ //

Hence, Proved

3. The vertices of odd degree (odd vertices) in a graph is always even.

→ • Let $G = (V, E)$ be a graph and V_e & V_o denotes even and odd vertices in G. then $V_e \subseteq V$, $V_o \subseteq V$ and $V_e \cup V_o = V$ and $V_e \cap V_o = \phi$

• $\sum_{V \in V} d(V) = \sum_{V_o \in V} d(V_o) + \sum_{V_e \in V} d(V_e)$

$2|E| = 2K + \sum_{V_o \in V} d(V_o)$

$\sum_{V_o \in V} d(V_o) = 2(|E| - K) =$ even no.

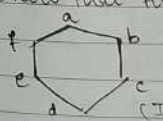
The each term $d(V)$ in the LHS should be even to make the sum of degrees and even no. hence the no. of vertices of odd degree must be even.

*** • Homomorphic Graph & Isomorphic Graph.

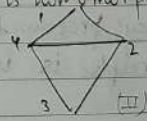
• Let $G = (V, E)$ and $G' = (V', E')$ be any two graph then a graph homomorphism f from $G \to G'$ is denoted by $\boxed{f : G \to G'}$

• It's a mapping f from $V \to V'$ from the vertex set of G to G' such that $(u, v) \in E \Rightarrow (f(u), f(v)) \in E'$

Q. Show that the below graph is homomorphic.



(I) $G = (V, E)$

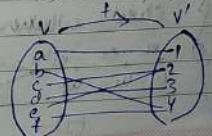


(II) $G' = (V', E')$

* let $f(a) = 1$ $f(d) = 3$

$f(b) = 4$ $f(e) = 2$

$f(c) = 2$. $f(f) = 4$



$(f,e) \in E$ in $G \Rightarrow (f(f), f(e)) \in E'$
$(1,2)$

$(c,d) \in E$ in $G \Rightarrow (f(c), f(d)) \in E'$ in G'.
$(2,3)$

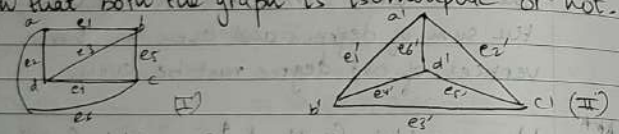
The graph is homomorphic.

- Two Graph $G = (V,E)$ & $G' = (V', E')$ are called isomorphic to each other if there's one to one corresponding between their vertices & edges such that incidence relationship is preserved (no. of pendant vertex connected is same).

① $|V| = |V'|$ ② $|E| = |E'|$ ③ $d(V) = d(V')$

④ Incidence Relationship is preserved.

- Show that both the graph is isomorphic or not.



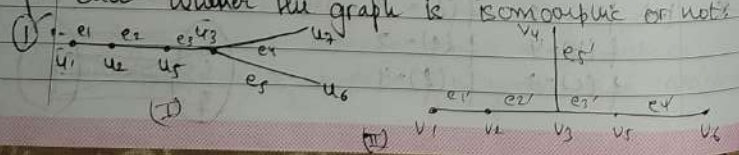
① Vertex Correspondance
$a \leftrightarrow a'$, $b \leftrightarrow b'$
$c \leftrightarrow c'$, $d \leftrightarrow d'$

② Edge Correspondance
$e1 \leftrightarrow e1'$, $e4 \leftrightarrow e4'$
$e2 \leftrightarrow e2'$ $e5 \leftrightarrow e5'$
$e3 \leftrightarrow e3'$ $e6' \leftrightarrow e6'$

③ $d(a) = d(a') = 3$, $d(c) = d(c') = 3$
$d(b) = d(b') = 3$. $d(d) = d(d') = 3$

④ Dependancy preserved.
Hence Graph is isomorphic.

Q. Check whether the graph is isomorphic or not?



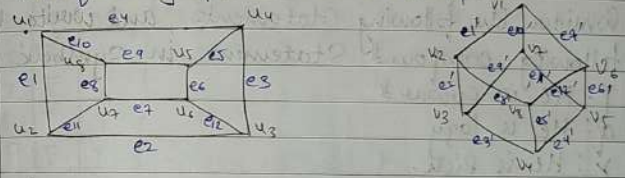
1. Vertex Correspondance
$u_1 \leftrightarrow v_1$ $u_4 \leftrightarrow v_4$
$u_2 \leftrightarrow v_2$ $u_5 \leftrightarrow v_5$
$u_3 \leftrightarrow v_3$ $u_6 \leftrightarrow v_6$

2. Edge Correspondance
$e_1 \leftrightarrow e_1'$ $e_4 \leftrightarrow e_4'$
$e_2 \leftrightarrow e_2'$ $e_5 \leftrightarrow e_5'$
$e_3 \leftrightarrow e_3'$

3. degree: $d(u_1) = d(v_1) = 1$ $d(u_4) = d(v_4) = 1$
$d(u_2) = d(v_2) = 2$ $d(u_5) = d(v_5) = 2$
$d(u_3) = d(v_3) = 3$ $d(u_6) = d(v_6) = 1$.

4. pendant vertex: $u_1 = 0$, $v_1 = 0$
$u_2 = 1$, $v_2 = 1$
$\boxed{u_3 = 2,\ v_3 = 1}$ — violated.

No Dependancy Preserved.
Therefore graph is not isomorphic.

② 

1. Vertex Correspondance
$u_1 \leftrightarrow v_1$, $u_5 \leftrightarrow v_5$
$u_2 \leftrightarrow v_2$, $u_6 \leftrightarrow v_6$
$u_3 \leftrightarrow v_3$, $u_7 \leftrightarrow v_7$
$u_4 \leftrightarrow v_5$, $u_8 \leftrightarrow v_8$

2. Edge Correspondance
$e_1 \leftrightarrow e_1'$, $e_5 \leftrightarrow e_5'$
$e_2 \leftrightarrow e_2'$, $e_6 \leftrightarrow e_6'$
$e_3 \leftrightarrow e_3'$, $e_7 \leftrightarrow e_7'$
$e_4 \leftrightarrow e_4'$, $e_8 \leftrightarrow e_8'$

3. degree: $d(u_1) = d(v_1) = 3$ $d(u_5) = d(v_5) = 3$
$d(u_2) = d(v_2) = 3$ $d(u_6) = d(v_6) = 3$
$d(u_3) = d(v_3) = 3$ $d(u_7) = d(v_7) = 3$
$d(u_4) = d(v_4) = 3$ $d(u_8) = d(v_8) = 3$

4. Dependency preservation:- $u_1 = v_1 = 0$, $u_2 = v_2 = 0$, $u_3 = v_3 = 0$, $u_4 = v_4 = 0$, $u_5 = v_5 = 0$, $u_6 = v_6 = 0$, $u_7 = v_7 = 0$ $u_8 = v_8 = 0$. Dependency preserved.
∴ Graph is isomorphic.